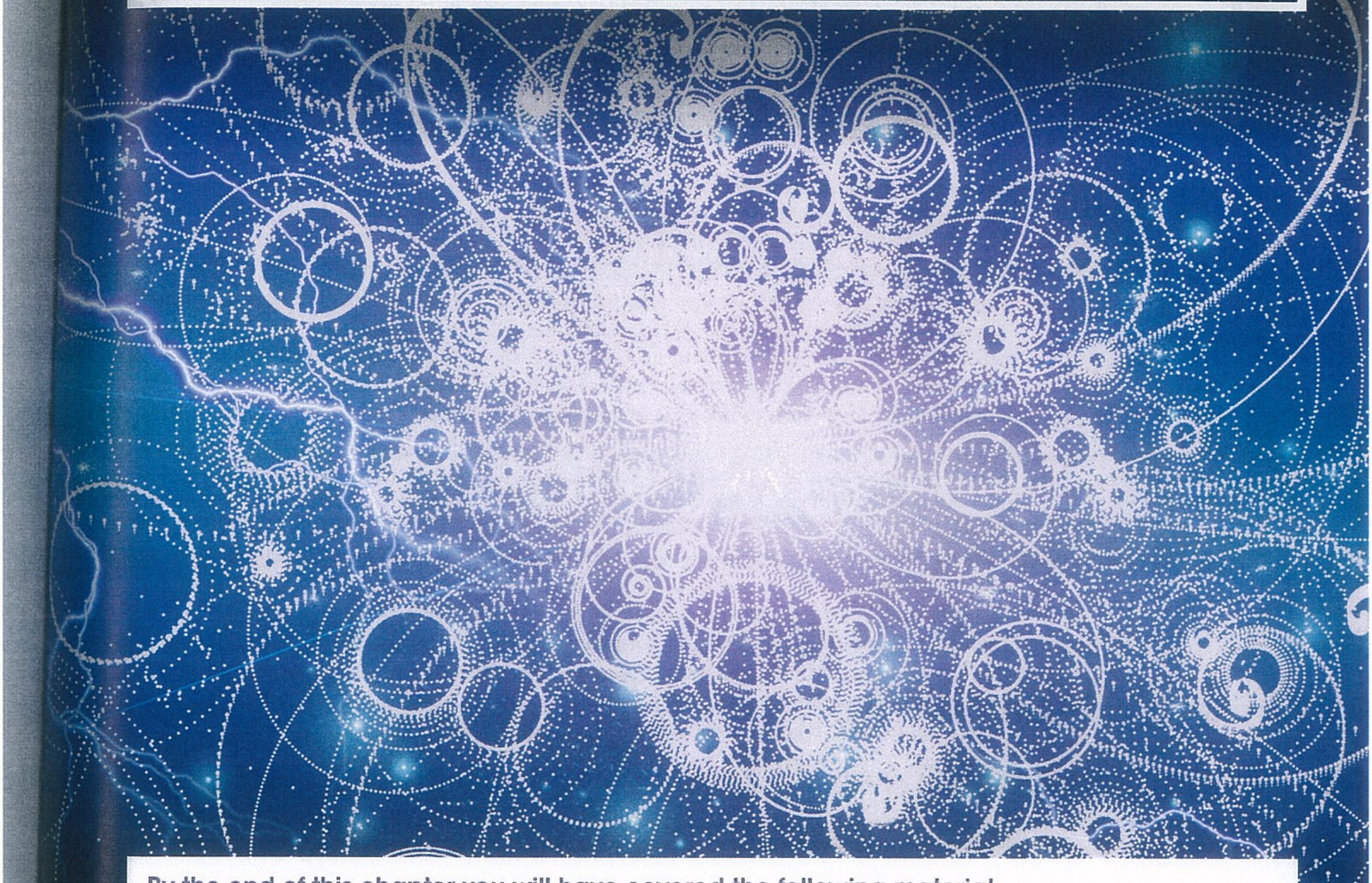
CHAPTER 10
THE STANDARD MODEL



By the end of this chapter you will have covered the following material.

Science Understanding

- The Standard Model is based on the premise that all matter in the universe is made up from elementary matter particles called quarks and leptons; quarks experience the strong force, leptons do not (ACSPH141)
- The Standard Model explains three of the four fundamental forces (strong, weak and electromagnetic forces) in terms of an exchange of force-carrying particles called gauge bosons; each force is mediated by a different type of gauge boson (ACSPH142)
- High-energy particle accelerators are used to test theories of particle physics including the Standard Model (ACSPH146)
- The Standard Model is used to describe the evolution of forces and the creation of matter in the Big Bang theory (ACSPH147)

Introduction

In the previous chapter we described the discovery of several new particles. We also looked at the evidence that indicates that familiar particles such as protons and neutrons are *not* elementary. This evidence includes the magnetic moment of uncharged particles such as neutrons and some other baryons.

The large number of new particles, more than 300, also suggested to physicists that not all of them could be elementary particles. This was partly because many of the particles seemed to come in 'families' or groups that had similar characteristics. For example, we have already seen that the proton and neutron have similar masses and the same spin. The group of three particles known as pions have almost identical masses and the same spin, but have different electric charges ($-1e$, 0 and $+1e$). As more and more particles were discovered, more patterns such as these began to emerge.

Other puzzles in particle physics included the nature of the field particles – photons, W and Z bosons, pions and the proposed gravitons. All but the last of these particles have now been detected. Photons are massless, but W and Z bosons and pions are not. What is the reason for the different masses of the field particles?

Physicists believed that there must be an underlying structure giving rise to the patterns observed in the properties and behaviour of all these particles. Just as the theory of evolution gave an underlying structure to the relationships between species, and the early theories of atoms gave an explanation for the organisation of the elements in the periodic table, physicists were searching for the underlying pattern to explain the 'particle zoo'. They wanted to know which of these particles were really elementary, and what the others were made of.

The answers to some of these questions came with the development of the **Standard Model** of particle physics. The first major development leading to the Standard Model was the **quark** theory.

QUARKS

Read about how the quark got its name.

At the same time that particle physicists were searching for the underlying structure of particles, astronomers and cosmologists were trying to understand how the large-scale structure of the universe came about. Astronomical evidence such as the red shift in the spectra of stars led to the development of the **Big Bang theory** – the theory that the universe began as a point-like **singularity** with enormous energy density. It may seem that the structure of the universe and the structure of a subatomic particle may not be closely related. However, according to the Big Bang theory, the universe began with the creation of elementary particles that condensed into the many particles we know of today, then into atoms, molecules and, finally, into the planets, stars and galaxies that give the universe its large-scale structure. Astronomical observations can only give us limited information about the history of the universe. To model what happened in the first stages of the evolution of the universe we need to use high-energy particle physics experiments.

Hence, an understanding of the underlying structure of the many particles in the particle zoo will also tell us about how the universe formed and then evolved to its current state. Once we understand the past evolution of our universe, we may be able to predict its future evolution.

The Standard Model of particle physics explains the underlying structure of particles and the forces via which they interact. It also describes the early stages of the evolution of the universe when particles first formed.

The evolution of the universe from a point-like singularity with enormous energy to the configuration of galaxies we know today is described by the Big Bang theory. The Big Bang theory relies on the Standard Model to describe and explain the first stages of the formation of the universe.

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Matter and the Standard Model

As we have seen, in the 1950s there were a huge number of particles known. Broadly, these could be classified as particles with mass – leptons and hadrons – and those without mass, which include exchange particles such as photons.

The hadrons did not appear to be fundamental particles. So the question was, what were they made from?

Gell-Mann invented the term 'quark' for these elementary particles. The spelling was borrowed from a rhyme in James Joyce's Finnegans Wake*: 'Three quarks for Muster Mark'. Quark is also a type of cottage cheese.*

Quarks

In 1963, Murray Gell-Mann and George Zweig independently proposed a model for the substructure of hadrons. According to their model, all hadrons are composed of two or three elementary constituents called quarks.

This early quark model had three types of quarks, designated by the symbols u, d and s, which are given the arbitrary names up, down and strange. The various types of quarks are called **flavours**. Figure 10.1 is a pictorial representation of the quark compositions of several hadrons.

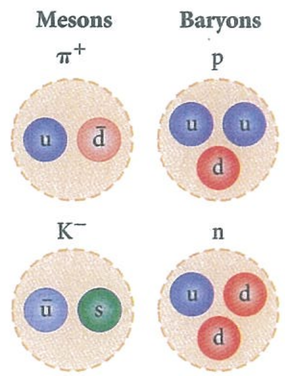


Figure 10.1 ▲ The quark composition of two mesons and two baryons

The compositions of all hadrons known when Gell-Mann and Zweig presented their model can be completely specified by three simple rules:

- A meson consists of one quark and one antiquark.
- A baryon consists of three quarks.
- An antibaryon consists of three antiquarks.

WORKED EXAMPLE 10.1

Which of these quark combinations is not a possible particle? (4 marks)

a $d\bar{d}$ **b** dd

c $\bar{u}\bar{d}d$ **d** $\bar{u}\bar{d}\bar{d}$

Answers	Logic	
a Possible	A meson consists of a quark-antiquark pair. A baryon consists of three quarks, and antibaryon of three antiquarks.	1 mark
b Not possible		1 mark
c Not possible	Hence, **a** is a meson, **b** is impossible, **c** is impossible and **d** is an antibaryon.	1 mark
d Possible		1 mark

Try this yourself

Which of these quark combinations is not a possible particle? (4 marks)

a uuu

b uu

c $u\bar{u}$

d $u\bar{u}\bar{u}$

Table 10.1 gives the properties of the quarks. The charmed c, top t, and bottom b quarks are described below.

Table 10.1 Properties of quarks

Name	Symbol	Spin	Charge	Baryon number	Strangeness	Charm	Bottomness	Topness
Up	u	$\frac{1}{2}$	$+\frac{2}{3}e$	$\frac{1}{3}$	0	0	0	0
Down	d	$\frac{1}{2}$	$-\frac{1}{3}e$	$\frac{1}{3}$	0	0	0	0
Strange	s	$\frac{1}{2}$	$-\frac{1}{3}e$	$\frac{1}{3}$	−1	0	0	0
Charmed	c	$\frac{1}{2}$	$+\frac{2}{3}e$	$\frac{1}{3}$	0	+1	0	0
Bottom	b	$\frac{1}{2}$	$\frac{1}{3}e$	$\frac{1}{3}$	0	0	+1	0
Top	t	$\frac{1}{2}$	$+\frac{2}{3}e$	$\frac{1}{3}$	0	0	0	+1

Each quark has an associated antiquark of opposite charge, baryon number, strangeness, charm, topness and bottomness.

Unlike any particles we have seen before, quarks carry a fractional electric charge. The u, d and s quarks have charges of $\frac{2e}{3}$, $-\frac{e}{3}$ and $-\frac{e}{3}$ respectively, where e is the electron charge ($e = 1.60 \times 10^{-19}$ C). Quarks have spin $\frac{1}{2}$, which means that they are classified as fermions (see Chapter 9), and have their own intrinsic magnetic moment and, hence, magnetic field. When quarks combine to form a particle, the charge of the particle is the arithmetic sum of the charges of its quarks. The spin of the particle is the sum of the spins of its quarks. However, spin is a vector, so we need to be more careful in how we add the spins. Both the magnitude and the direction of spin are quantised.

Recall from Chapter 4 that magnetic moment is a vector. It points in the direction of the north pole of an equivalent current loop. The magnitude of the magnetic moment of a particle depends on the spin.

In Table 10.1 you can see that quarks have baryon number $\frac{1}{3}$. Antiquarks have baryon number $-\frac{1}{3}$. Hence, a meson composed of one quark and one antiquark has baryon number $B = \frac{1}{3} + \left(-\frac{1}{3}\right) = 0$. A baryon, which is composed of three quarks, has $B = \frac{1}{3} + \frac{1}{3} + \frac{1}{3} = 1$, and an antibaryon composed of three antiquarks has $B = \left(-\frac{1}{3}\right) + \left(-\frac{1}{3}\right) + \left(-\frac{1}{3}\right) = -1$.

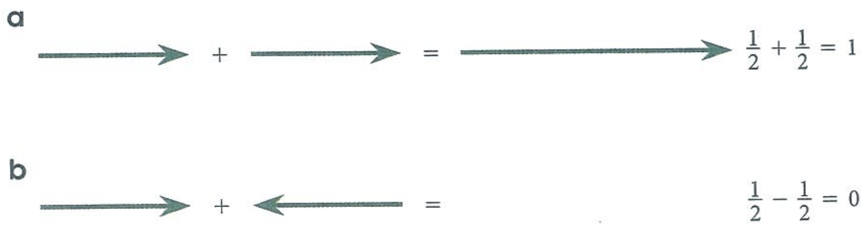
The law of conservation of baryon number, which we used in Chapter 9, is based on experimental observation. It is consistent with the model of quarks as elementary particles and this assignment of baryon numbers to quarks. Hence, the quark theory is able to explain the experimental observation of baryon number conservation.

Although no isolated quark has ever been experimentally observed, the quark model does describe the properties of mesons and baryons. Table 10.2 shows how mesons can be constructed from combinations of a quark and an antiquark.

Table 10.2 Quark composition of some mesons

		Antiquarks									
		$\bar{b}$		$\bar{c}$		$\bar{s}$		$\bar{d}$		$\bar{u}$	
Quarks	b	Υ	$(\bar{b}b)$	B_{c-}	$(\bar{c}b)$	$\bar{B}_s$	$(\bar{s}b)$	$\bar{B}_{d^0}$	$(\bar{d}b)$	B^-	$(\bar{u}b)$
	c	B_c^+	$(\bar{b}c)$	J/Ψ	$(\bar{c}c)$	D_s^+	$(\bar{s}c)$	D^+	$(\bar{d}c)$	D^0	$(\bar{u}c)$
	s	B_s^0	$(\bar{b}s)$	D_s-	$(\bar{c}s)$	ϕ	$(\bar{s}s)$	$\bar{K}_0$	$(\bar{d}s)$	K^-	$(\bar{u}s)$
	d	B_d^0	$(\bar{b}d)$	D^-	$(\bar{c}d)$	K^0	$(\bar{s}d)$	π^0	$(\bar{d}d)$	π^-	$(\bar{u}d)$
	u	B^+	$(\bar{b}u)$	$\bar{D}_0$	$(\bar{c}u)$	K^+	$(\bar{s}u)$	π^+	$(\bar{d}u)$	π^0	$(\bar{u}u)$

Consider the π^- meson as an example. We see in Table 10.2 that this meson has the quark composition $\bar{u}d$. The $\bar{u}$ quark has baryon number $-\frac{1}{3}$ and the d quark has baryon number $\frac{1}{3}$, giving a total of $B = 0$. The $\bar{u}$ quark has charge $-\frac{2e}{3}$ and the d quark has charge $-\frac{e}{3}$. This gives the π^- meson a total charge of $-1e$. Both the $\bar{u}$ and the d quarks have spin $\frac{1}{2}$. Spins are a little more complicated to add up because they represent magnetic moments, which are vectors. The magnetic moments can only line up in two ways: parallel or antiparallel (see Figure 10.2). When magnetic moments are parallel the total spin is $\frac{1}{2} + \frac{1}{2} = 1$; when they are antiparallel the total spin is $\frac{1}{2} - \frac{1}{2} = 0$. Looking back at Table 9.1 on page 264, we can see that a π^- meson has the properties $B = 0$, charge $= -1e$ and spin $= 0$. This is exactly what we expect from the combination of an anti-up quark and a down quark with their spins antiparallel. The alternative spin arrangement with spin $= 1$ corresponds to an excited (higher energy) state of the π^- meson.



◀ **Figure 10.2**
a) Parallel spins add to give spin = 1.
b) Antiparallel spins add to give spin = 0.

Baryons are composed of three quarks, and antibaryons of three antiquarks. Table 10.3 gives the quark composition of some baryons.

Note: Some baryons have the same quark composition, such as the p and Δ^+ and the n and Δ^0 baryons. In these cases, the Δ particles are considered to be excited states of the proton and neutron.

Table 10.3 Quark composition of several baryons

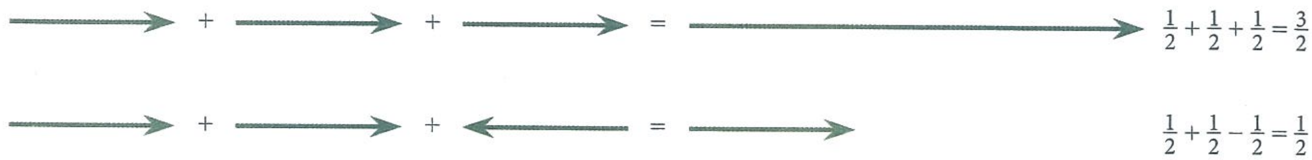
Particle	Quark composition
p	uud
n	udd
Λ^0	uds
Σ^+	uus
Σ^0	uds
Σ^-	dds
Δ^{++}	uuu
Δ^+	uud
Δ^0	udd
Δ^-	ddd
Ξ^0	uss
Ξ^-	dss
Ω^-	sss

Recall from Chapter 9 that particles with integer spin are bosons and particles with half-integer spin are fermions. The two types of particles behave differently when they interact and bind together. Fermions obey the exclusion principle; bosons do not.

WORKED EXAMPLE 10.2

Consider a baryon with quark composition uud. Show that it has the properties of a proton. (5 marks)

Answer	**Logic**	**Marks**
The uud combination has total charge $\frac{2}{3}e + \frac{2}{3}e - \frac{1}{3}e = 1e.$	An up quark has charge $\frac{2}{3}$ and a down quark has charge $-\frac{1}{3}$.	2 marks
Its baryon number is $B = \frac{1}{3} + \frac{1}{3} + \frac{1}{3} = 1$	Each quark has $B = \frac{1}{3}$.	1 mark
Spin $= \frac{3}{2}$ or $\frac{1}{2}$	The possible spin states are $\frac{3}{2}$ (all parallel) and $\frac{1}{2}$ (one antiparallel, two parallel), as shown in Figure 10.3.	2 marks



▲ **Figure 10.3** Possible spin arrangements for three quarks

Looking back at Table 9.1 in Chapter 9, we can see that the proton has $B = 1$, charge $= +1e$ and spin $= \frac{1}{2}$, in agreement with this quark composition with the lower energy arrangement of spins.

Try this yourself

Repeat the example above for a neutron with quark composition udd. (5 marks)

In the example above, the spin $\frac{3}{2}$ arrangement corresponds to a Δ^+ particle. As with the mesons, the parallel spin states are considered to be excited particle states. Hence, the Δ^+ particle is considered to be an excited state of the proton.

Although the original quark model was highly successful in classifying particles into families, some discrepancies occurred between its predictions and certain experimental decay rates. Consequently, several physicists proposed a fourth quark flavour in 1967. They argued that if four types of leptons existed (as was thought at the time), there should also be four flavours of quarks because of an underlying symmetry in nature. The fourth quark, designated c, was assigned a property called charm. A *charmed* quark has charge $+\frac{2e}{3}$, just as the up quark does, but its charm distinguishes it from the other three quarks. This introduces a new quantum number C, representing charm. The new quark has charm $C = +1$, its antiquark has charm of $C = -1$; all other quarks have $C = 0$.

At the time of the development of the quark theory only four types of lepton were known. We now know that there are six types, as described in Chapter 9.

Evidence that the charmed quark exists began to accumulate in the 1970s, when a series of heavy mesons with long lifetimes were discovered. The existence of these new mesons provided firm evidence for the fourth quark flavour.

Then in 1975, a fifth type of lepton, called the tau (τ) lepton, was discovered. This led physicists to propose that more flavours of quarks might exist, on the basis of symmetry arguments similar to those leading to the proposal of the charmed quark. These proposals led to more elaborate quark models and the prediction of two new quarks, top (t) and bottom (b). To distinguish these quarks from the others, quantum numbers called *topness* and *bottomness* (with allowed values +1, 0, −1) were assigned to all quarks and antiquarks (see Table 10.1).

ACTIVITY 10.1

MAKING BARYONS

Aim

Your aim is to use 'quarks' to make some 'baryons' and identify their properties.

You will need

At least four beads or tokens in each of six different colours. Each different colour represents a different type of quark. You will also need a bag to hold the quarks, coloured pencils and paper.

What to do

1. Make a table listing the types of quarks and their properties, including charge, spin, strangeness, charm, topness and bottomness. Use Table 10.1 as a guide and add a column listing the colour you are using to represent each quark.
2. Place all the beads or tokens in the bag and randomly take 3 quarks from the bag.
3. These three quarks make up your new baryon. Using the table you have already drawn, work out the properties of your new baryon. List charge, spin, strangeness, charm, topness and bottomness.

   Note that spins can be in the same direction or opposite direction. Hence, there is not a single unique value for the spin of your new baryon.
4. Use your coloured pencils to record your new baryon and write down its properties.
5. Now see whether you can identify it in a list of known baryons, such as those given in Table 10.3 and Table 9.1 (page 264).
6. Put the quarks back in their bag and repeat the process.

What did you discover?

1. How many different baryons do you think you can make?
2. Did you make the same baryon more than once? Do you think all quarks are as common in nature as they are in your bag? Why or why not?

Matter particles in the Standard Model

The Standard Model of particle physics says that all matter is made up of two types of elementary particle. These are quarks and leptons.

As we saw in Chapter 9, there are six types of leptons, each with its own antiparticle. These are the electron, muon, tau, electron neutrino, muon neutrino and tau neutrino, and the antiparticle of each. Leptons do not combine to form other particles.

There are also six types or flavours of quarks, as we have just seen: up, down, top, bottom, strange and charmed. Each type of quark also has an antiquark. Quarks bind together to form other particles. They bind in groups of three to form baryons and in quark–antiquark pairs to form mesons.

All quarks and leptons have spin of $\frac{1}{2}$, and have an intrinsic magnetic moment. They are all fermions. This means that when they bind together into other particles, including baryons or large composite particles such as atoms, they obey the exclusion principle.

The second part of the Standard Model that we shall explore is the interaction of these particles. What force binds quarks together to form baryons and mesons? How is this force related to the force that holds nucleons together in the nucleus? We shall answer these questions in the next section.

Figure 10.4 ▶ The elementary matter particles described in the Standard Model

Quarks		Leptons	
u	$\bar{u}$	e^-	e^+
d	$\bar{d}$	ν_e	$\bar{\nu}_e$
s	$\bar{s}$	μ^-	μ^+
c	$\bar{c}$	ν_μ	$\bar{\nu}_\mu$
t	$\bar{t}$	τ^-	τ^+
b	$\bar{b}$	ν_τ	$\bar{\nu}_\tau$

QUESTION SET 10.1

Remembering

1 Name the three types of elementary particles.

2 a How many quarks do mesons consist of?

  b How many quarks do baryons consist of?

Understanding

3 Explain why particle physics experiments are important in the development of cosmological theories such as the Big Bang theory.

4 Which of these quark combinations is a possible particle? Explain your answers.

  a ud

  b udd

  c $\bar{u}dd$

  d $\bar{u}\bar{d}\bar{d}$

  e $u\bar{d}$

Applying

5 For each of these quark combinations, give the resulting particle's charge, strangeness and baryon number and all possible spins.

  a $\bar{d}\bar{d}\bar{s}$

  b bbt

  c uus

  d $\bar{u}\bar{u}\bar{s}$

6 Show that the properties of the quark combination $\bar{s}u$ are consistent with the K^+ meson.

7 Show that the properties of the quark combination uds are consistent with the Λ^0. What other baryon could be composed of these quarks? Explain your answer.

Analysing

8 A proton (uud) and an antiproton ($\bar{u}\bar{u}\bar{d}$) collide and a u and a $\bar{u}$ annihilate to create a total of eight quarks: ($u\,\bar{u}\,d\,\bar{d}\,d\,\bar{d}\,s\,\bar{s}$). A $\bar{\Sigma}^0$, a Λ^0 and what other particle are formed?

9 Analyse each of the following reactions in terms of constituent quarks and show that each type of quark is conserved.

a $\pi^+ + p \rightarrow K^+ + \Sigma^+$

b $K^- + p \rightarrow K^+ + K^o + \Omega^-$

10 Compare and contrast the aims of particle physics and cosmology. How has your understanding of the interaction between theories and researchers in different areas of physics changed?

Forces and the Standard Model

So far we have looked at four forces: the electromagnetic, gravitational, strong nuclear and weak nuclear forces. We said in the previous chapter that three of these, the electromagnetic, gravitational and weak nuclear, are fundamental forces. The fourth, the strong nuclear force, is not a fundamental force. Instead it is a consequence of another force, the strong force, which we shall introduce in this section.

The quark theory, combined with a theory called gauge theory, allows us to understand why the strong nuclear force is not a fundamental force. Rather, it arises from the interactions of quarks, as the nucleons (protons and neutrons) are composed of quarks. Gauge theory is highly mathematical and well beyond the scope of this course, so we shall look at some of the predictions of the theory only.

Gauge theory predicts the existence of particles called **gauge bosons**, which are massless particles via which massive particles interact.

We have already met one of these gauge bosons several times before: the photon.

The fundamental forces and field particles

In the Standard Model, particle interactions are described in terms of field particles or exchange particles. The emission of a field particle by one particle and its absorption by another particle manifests as a force between the two interacting particles.

Physicists now believe there are four fundamental forces which explain all interactions between particles.

In order of decreasing strength, they are the strong force, the electromagnetic force, the weak force and the gravitational force.

The strong force is responsible for the attractive force between nucleons. It has a very short range and is negligible for separation distances between nucleons greater than approximately 10^{-15} m (about the size of the nucleus). It acts between quarks, and the exchange or field particle is called the gluon. The name 'gluon' comes from the way that gluons 'glue' the nucleus together.

The electromagnetic force, which binds atoms and molecules together to form ordinary matter, has a strength of approximately 10^{-2} times that of the strong force. As we have seen, this long-range force decreases in magnitude as the inverse square of the separation between interacting particles. The field particles for the electromagnetic force are photons.

Do not confuse the strong force with the strong nuclear force. They are different forces, acting between different particles and mediated by different exchange particles.

The weak force is a short-range force that is responsible for beta decay and other decay processes. Its strength is only about 10^{-5} times that of the strong force. The weak force is mediated by field particles called *W* and *Z* bosons.

Finally, the gravitational force is a long-range force that has a strength of only about 10^{-39} times that of the strong force. Although this is the force that holds the planets, stars and galaxies together, its effect on elementary particles is negligible. The gravitational force is thought to be mediated by field particles called gravitons. The gravitational force is not part of the Standard Model, and the graviton has not yet been detected.

These interactions, their ranges, and their relative strengths are summarised in Table 10.4.

You can see that these forces differ in several respects, including their relative strength and the types of particle they act on.

Table 10.4 The fundamental forces and their exchange or field particles

Interactions	Relative strength	Range of force	Mediating field particle	Mass of field particle (GeV c^{-2})
Strong	1	Short (≈1 fm)	Gluon	0
Electromagnetic	10^{-2}	∞	Photon	0
Weak	10^{-5}	Short (≈10^{-3} fm)	$W^{\pm}$, Z^0 bosons	80.4, 80.4, 91.2
Gravitational	10^{-39}	∞	Graviton	0

Relative strengths of forces

The relative strength of a force is a way of describing how strong the interaction is that it causes between particles. As an illustration, let's consider the forces between two electrons. We already know from Chapter 3 that electrons interact via the electromagnetic force. The electromagnetic force exerted by one electron on another is described by Coulomb's law:

$$F_{\text{ee, EM}} = \frac{k_e q_e^2}{r^2} = \frac{2.307 \times 10^{-28}}{r^2}\ \text{N m}^{-2}$$

But electrons have mass, and so as well as exerting an electrostatic force, they will also exert a *gravitational* force on each other:

$$F_{\text{ee, G}} = \frac{G m_e^2}{r^2} = \frac{5.535 \times 10^{-71}}{r^2}\ \text{N m}^{-2}$$

So we don't worry about the gravitational force between two electrons because the electromagnetic force is much stronger than the gravitational force.

As we can see from the equations, the exact magnitude of the forces depends on the properties of the particles that are interacting (charge for electromagnetism and mass for gravity), the separation between them, and factors relating to the force itself. These factors are represented by the constants k_e and G in the equations above. To compare the strengths of the fundamental forces alone, independently of properties such as mass and charge, physicists have introduced dimensionless quantities called coupling constants or relative strengths, given in Table 10.4. The exact definition of these constants depend on some quite complex theory.

The strong force acts between quarks and is mediated by gluons. The strong nuclear force acts between nucleons and is mediated by pions. Other sources may refer to the strong nuclear force as the strong force. Make sure you know which is meant.

The strong force and the strong nuclear force

The nuclear force was historically called the strong nuclear force. Once the quark theory was established, however, the phrase 'strong force' was reserved for the force between quarks. We shall follow this convention: the strong force is between quarks or particles built from quarks, and the nuclear force is between nucleons in a nucleus. The nuclear force is a result of the strong force. Other books and resources may refer to the nuclear force as the strong force.

The Standard Model, in which quarks interact via gluons, gives us a different way of looking at nuclear interactions. The proton–neutron interaction was described in Chapter 9 as being the result of an exchange of a pion between the two particles (see Figure 9.8, page 268).

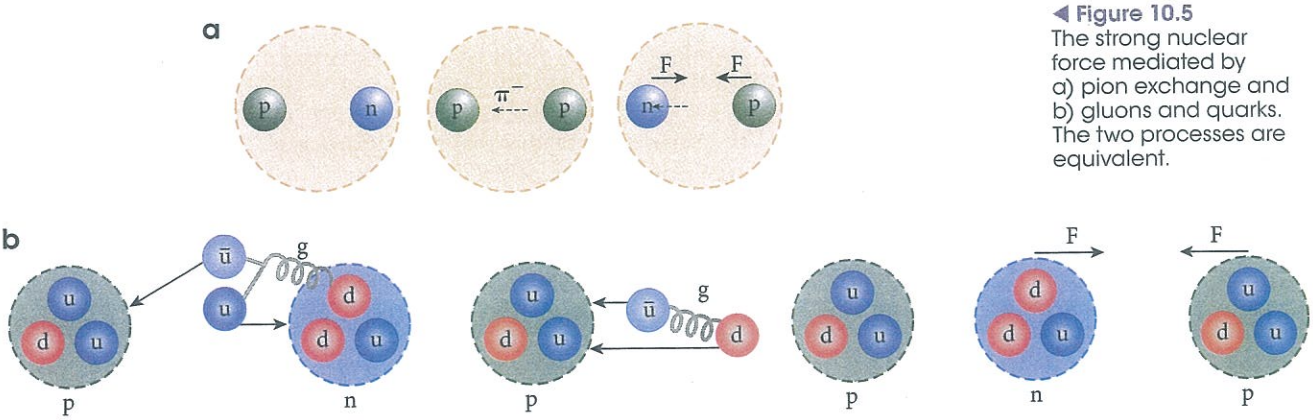
Let's look at the same interaction from the viewpoint of the quark model, shown in Figure 10.5. Figure 10.5(a) shows the same process as Figure 9.8, although we now represent the process in three steps. First, the emission of a π^- particle by the neutron. The emission of a π^- changes the neutron to a proton. The absorption of the π^- by the proton results in it being converted into a neutron. The net effect is a force acting between the two nucleons. In Figure 10.5(b), the proton and neutron are represented by their quark constituents. Each quark in the neutron and proton is continuously emitting and absorbing gluons. The energy of a gluon can result in the creation of quark–antiquark pairs. This process is similar to the creation of electron–positron pairs in pair production. When the neutron and proton approach to within 1 fm of each other, these gluons and quarks can be exchanged between the two nucleons, and such exchanges produce the nuclear force.

Figure 10.5(b) shows one possibility for the process shown in Figure 10.5(a). A down quark in the neutron on the right emits a gluon. The energy of the gluon creates a $u\bar{u}$ pair. The u quark stays within the neutron (which has now changed to a proton), and the recoiling d quark and the $\bar{u}$ antiquark are transmitted to the proton. Here the $\bar{u}$ annihilates a u quark within the proton and the d is captured. The net effect is to change a u quark to a d quark, and the proton on the left has changed to a neutron.

As the d quark and $\bar{u}$ antiquark in Figure 10.5(b) transfer between the nucleons, the d and $\bar{u}$ exchange gluons with each other and hence are bound to each other by means of the strong force.

Looking back at Table 10.2, we see that this combination is a π^-, or Yukawa's field particle, as described in Chapter 9. Therefore, the quark model of interactions between nucleons is consistent with the pion-exchange model.

The pion was introduced in Chapter 9 as the field particle that mediates the strong nuclear force.



◀ Figure 10.5 The strong nuclear force mediated by a) pion exchange and b) gluons and quarks. The two processes are equivalent.

The electroweak force

In 1979 Sheldon Glashow, Abdus Salam and Steven Weinberg won the Nobel Prize in Physics for developing a theory that unifies the electromagnetic and weak interactions. This **electroweak theory** postulates that the weak and electromagnetic interactions have the same strength when the particles involved have very high energies. Because of the mass difference between photons and the W and Z bosons, the electromagnetic and weak forces are quite distinct at low energies, but become similar at very high energies, when the rest energy is negligible relative to the total energy. This behaviour, as one goes from high to low energies, is called *symmetry breaking* because the forces are similar, or symmetric, at high energies but are very different at low energies.

The two interactions are viewed as different manifestations of a single unifying electroweak interaction. The theory makes many concrete predictions, but perhaps the most spectacular is the prediction of the masses of the W and Z particles at approximately 82 GeV c^{-2} and 93 GeV c^{-2}, respectively. These predictions are close to the masses in Table 10.4 determined by experiment.

The electroweak theory is an example of a unifying theory. Maxwell's theory of electromagnetism is another example of a unifying theory. Unifying or unified theories are those that explain more than one force within a single theoretical framework.

We shall see that this separation of the electromagnetic and weak forces was an important step in the evolution of the universe. The electroweak theory was important in the development of the Big Bang theory.

Mass and the Higgs boson

One of the questions raised by the Standard Model is why all the field particles except the W and Z bosons are massless. Or, put another way, why do the W and Z bosons have mass, and in fact what is the origin of the mass of all the other massive elementary particles?

To resolve this problem, a hypothetical particle called the Higgs boson, which provides a mechanism for breaking the electroweak symmetry, has been proposed. When particles interact via the Higg's field, they gain mass from this interaction. The field particle is the Higgs boson. The Standard Model modified to include the Higgs boson provides a logically consistent explanation of the massive nature of the W and Z bosons.

The Higgs boson was named for Peter Higgs, one of a group of physicists who proposed its existence in 1964. The existence of the Higgs boson was confirmed in March 2013 by CERN (the European Organization for Nuclear Research) after a particle with properties corresponding to those predicted for the Higgs boson was observed in July 2012, as described in the Scientific literacy box on the next page.

The Standard Model: a summary

Particle physicists now believe that all matter is made up of two types of elementary particles: quarks and leptons. There appears to be a symmetry between the quarks and leptons, in that there are six types of each, along with their antiparticles.

Matter is made up of the elementary particles, quarks and leptons. Interactions between these particles are mediated by the third type of elementary particle, gauge bosons or field particles.

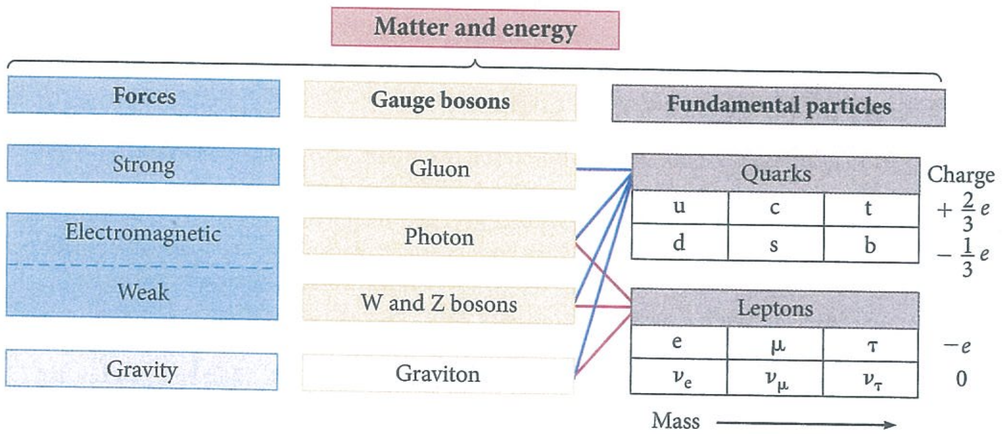
Quarks combine to form mesons and hadrons. Mesons are made up of two quarks, as shown in Table 10.2. Baryons are made up of three quarks each (see Table 10.3).

In addition to these particles, there are also the exchange particles associated with the four fundamental forces, as shown in Table 10.4.

Although the details of the Standard Model are complex, its essential ingredients are summarised in Figure 10.6. This diagram shows that quarks participate in all the fundamental forces and that leptons participate in all except the strong force.

Note that the Standard Model *does not* include the gravitational force at present. However, we include gravity in Figure 10.6 because physicists hope to eventually incorporate this force into a **unified theory**.

Figure 10.6 ▶ The Standard Model of particle physics



8 Use the information on black body radiation in Chapter 7 to find the peak wavelength radiated by:
   a a black body at 3000 K (the universe at 700 000 years after the Big Bang).
   b a black body at 3 K (the universe today).

Analysing

9 The hydrogen emission spectrum has a series of visible lines called the Balmer series, as described in Chapter 8. This series corresponds to transitions to the $n = 2$ energy level. For a distant galaxy, the line corresponding to the $n = 4$ to $n = 2$ transition is observed at a wavelength of 700 nm.
   a Find the wavelength of the line corresponding to the $n = 4$ to $n = 2$ transition on Earth.
   b Find the speed at which this galaxy is receding from us.
   c Find the distance of this star from us.

Reflecting

10 Draw a concept map to illustrate your understanding of the Big Bang theory. Include the evidence for the theory, as well the limitations of the theory, on your diagram. Include the things that the theory predicts and explains. Add links showing how it draws on ideas from other areas of physics.

CHAPTER SUMMARY

- The Standard Model of particle physics is the currently accepted model that describes the elementary particles of which matter is made and explains the forces via which they interact.
- The Big Bang theory is currently the most widely accepted cosmological model for the evolution of the universe, beginning with a singularity around 14 billion years ago. It relies on particle physics to explain the evolution of matter and forces from this initial singularity.
- More than 300 distinct types of particles have been observed. These can be classified as leptons and hadrons. Hadrons can be further classified as mesons and baryons.
- The Standard Model says that all hadrons are composed of elementary particles called quarks.
- Quarks come in six 'flavours': up, down, top, bottom, charmed and strange. For each quark type there is an antiquark.
- Mesons are composed of one quark and one antiquark.
- Baryons are composed of three quarks. Their corresponding antiparticles are composed of the corresponding three antiquarks.
- Interactions between particles occur via the exchange of field particles. According to the Standard Model of particle physics, these field particles are photons for the electromagnetic force, W and Z bosons for the weak force and gluons for the strong force.
- The four fundamental forces are the strong force (not the same as the strong nuclear force), the weak force, the electromagnetic force and the gravitational force.
- The strong nuclear force, which is mediated by pions, is a result of the strong force interaction between quarks, which is mediated by gluons.
- The Standard Model does not currently include gravitation, although some theories predict the existence of the graviton as the exchange particle for the gravitational force, and the existence of gravitational waves (analogous to photons and electromagnetic waves).
- In the Standard Model, particles acquire mass via interactions with the Higgs field. The mediating particle is the Higgs boson.
- The Higgs boson was first observed at CERN in July 2012.
- The Standard Model has limitations. It does not include a theory for gravity, and it predicts that neutrinos should be massless. Experimentally, neutrinos have been found to have very small but non-zero mass.